

Assumption (thought)	P (prior)	P (after)	KLD
wantsNotDrink(mary,coffee):	0.40	0.9208	0.1872
wantsNotDrink(mary,coke):	0.50	0.92	0.1151
wantsNotDrink(mary,fruitTea):	0.60	0.0199	1.4636
wantsNotDrink(mary,peppermintTea):	0.60	0.4	0.0433
wantsNotDrink(mary,proteinShake):	0.80	0.9831	0.0182
wantsNotDrink(mary,redBull):	0.80	0.98	0.0176
wantsNotDrink(mary,tea):	0.50	0.88	0.0973
wantsNotDrink(mary,water):	0.50	0.1	0.4047
			2.3472

We can see that not only the proposition that MARY DOES NOT WANT TO DRINK COFFEE becomes much more manifest to Peter, but also the propositions to the effect that Mary wouldn't want similar tiredness-blocking drinks such as coke and Red Bull. Moreover, the manifestness of the assumptions that Mary is unlikely to drink fruit tea or peppermint tea are significantly decreased. In other words, we can see that Mary's utterance not only conveys a strong implicature about Mary not wanting to drink coffee, but also a variety of more weakly communicated implicatures.

In our model we treat this interpretive hypothesis as the speaker-intended one. We notice that in this interpretation, the presumed relevance expectations are satisfied in the sense that inferences changing the probability distribution of the queries of interest are available, and the knowledge base providing access to these inferences is not bloated, i.e. there are little or no superfluous assumptions in the knowledge graph.

5.4.2 Interpretive hypothesis Int 2

This interpretive hypothesis is chosen to illustrate what happens if we construct a hypothesis that does not satisfy the extant relevance expectations. To this end we cut the knowledge graph representing encyclopaedic knowledge to cover only assumptions about coffee, the only beverage mentioned directly in Peter's question.

Assumption (thought)	P (prior)	P (after)	KLD
wantsNotDrink(mary,coffee):	0.40	0.9208	0.1873
wantsNotDrink(mary,coke):	0.50	0	inf
wantsNotDrink(mary,fruitTea):	0.60	0	inf
wantsNotDrink(mary,peppermintTea):	0.60	0	inf
wantsNotDrink(mary,proteinShake):	0.80	0	inf
wantsNotDrink(mary,redBull):	0.80	0	inf
wantsNotDrink(mary,tea):	0.50	0	inf
wantsNotDrink(mary,water):	0.50	0	inf
			inf

Notice that in this case, many queries of interest cannot be answered, leading to meaningless values of the KL divergence, which is set to infinity.

5.4.3 Interpretive hypothesis Int 3

This interpretive hypothesis involves a more elaborate knowledge graph of encyclopaedic knowledge about energizing drinks and drinks offered at parties in the sense

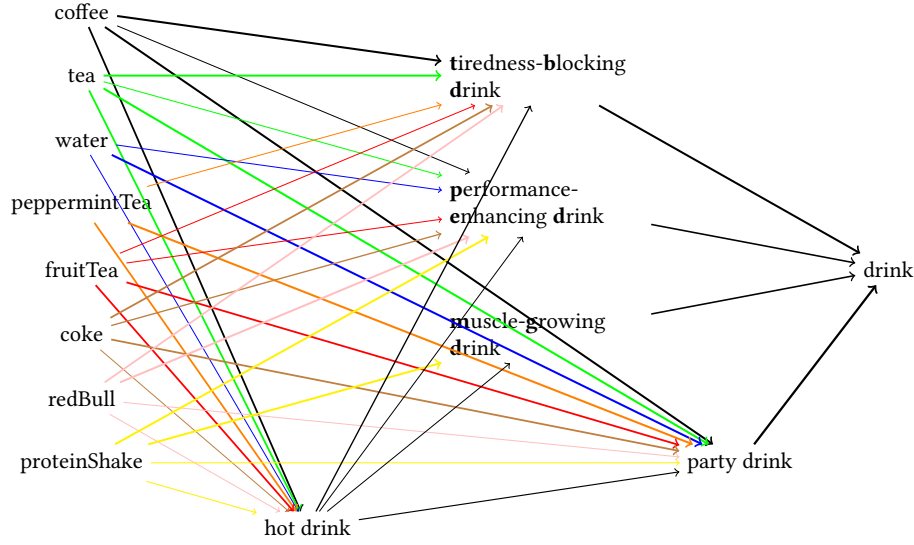


Figure 4: Elaborate knowledge graph of encyclopaedic knowledge about energizing drinks and drinks offered at parties.

that it adds a layer of classification into hot drinks (and, in consequence, cold drinks) along the path from individual beverages to the final category. (See Figure 4).

However, this additional layer of encyclopaedic knowledge is not needed to satisfy relevance expectations. In other words, it is an interpretation that is more costly to construct than Int 1.

Assumption (thought)	P (prior)	P (after)	KLD
wantsNotDrink(mary,coffee):	0.40	0.97503616	0.2186
wantsNotDrink(mary,coke):	0.50	0.926848	0.1183
wantsNotDrink(mary,fruitTea):	0.60	0.69107248	0.0063
wantsNotDrink(mary,peppermintTea):	0.60	0.81088	0.0302
wantsNotDrink(mary,proteinShake):	0.80	0.98599328	0.0188
wantsNotDrink(mary,redBull):	0.80	0.981712	0.0180
wantsNotDrink(mary,tea):	0.50	0.962176	0.1349
wantsNotDrink(mary,water):	0.50	0.17704	0.1962
0.7411			

In the results we can see that the stronger and weaker implicatures can be derived in similar ways to Int 1. Yet the utility of the utterance as indicated by the KL divergence between $P(\text{prior})$ and $P(\text{after})$ is lower in Int 3 than in Int 1 due to the additional probability calculations about the additional edges in the graph.

In classical relevance theory, Int 3 is said to be less relevant than Int 1 because the processing effort involved exceeds that involved in processing Int 1, and this effort is not offset by additional cognitive effects contributing to the satisfaction of relevance expectations. In the probabilistic computational model we are developing, this difference is captured by the lower value of the sum total of the KL divergence. However, more costly to construct interpretive hypotheses do not always affect the KL measure, as we shall see directly when we turn to discuss yet another interpretive

hypothesis.

5.4.4 Interpretive hypothesis Int 4

Int 4 includes the same encyclopaedic knowledge graph about beverages as in Int 1. In addition to this it provides another graph that allows to draw conclusions about whether Mary, if she does not want to drink coffee after 6PM, may be a particular health-conscious person. This second graph does not affect the probability calculations involving the edges of the graph describing the encyclopaedic knowledge about beverages.

However, if we assume that Peter's relevance expectations are the same as in Int 1, then the additional possible inferences do not affect the answers available to the queries corresponding the relevance expectations:

Assumption (thought)	P (prior)	P (after)	KLD
wantsNotDrink(mary,coffee):	0.40	0.9208	0.1873
wantsNotDrink(mary,coke):	0.50	0.92	0.1151
wantsNotDrink(mary,fruitTea):	0.60	0.0199	1.4636
wantsNotDrink(mary,peppermintTea):	0.60	0.4	0.0433
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wantsNotDrink(mary,tea):	0.50	0.88	0.0973
wantsNotDrink(mary,water):	0.50	0.1	0.4047
			2.3472

The KL measure of this interpretation Int 4 is the same as that of Int 1. Int 4 and Int 1 differ in another dimension: Int 4 has more so-called *atoms* in the ProbLog program. In logic programming in languages such as Prolog and ProbLog, an *atom* is essentially a predicate representing a proposition that is assumed to hold true (with a certain probability) in the model of the world that the program is representing. The set of such atoms in a ProbLog program is called its *Herbrand base* (Flach 1994, Section 2.1). Since Int 4 has an additional encyclopaedic knowledge graph over against Int 1, its Herbrand base is larger. It is natural to assume that a larger Herbrand base means increased processing effort for processing this hypothesis.

Thus we have another way to differentiate two interpretive hypotheses according to the processing effort they incur: when the respective hypotheses do not differ in their KL measure, they will differ in the size of their Herbrand base.

5.4.5 Interpretive hypothesis Int 5

Int 5 contains the same knowledge base as Int 4. This time, we assume that information about Mary's being a particular health-conscious person is relevant to Peter if available (perhaps for drawing further conclusions not just about beverages Mary may or may not accept but also about what sort of foods or snacks Mary may appreciate at the party). Thus, the additional knowledge graph contributes additional possible inferences satisfying Peter's relevance expectations. Consequently, this interpretive hypothesis is more relevant to Peter than Int 1, as the increased processing effort involved in utilising the additional contextual knowledge satisfies more relevance expectations. This is reflected in a greater KL measure than the one in Int 1.